

Focus, Passionate & critical thinking Assessment

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Section A: Workplace Scenario Analysis

Scenario:

You've recently joined ABC Tech, a software development company, as a junior developer. As part of your role, you've been assigned to a project team responsible for maintaining and improving the company's website. Today, you encountered performance issues with the website, like

- High server loads during peak hours, slow response times, and intermittent downtimes.
- Slow data retrieval and increased server load due to unoptimized database queries
- Rendering problems and decreased performance on certain browsers

Your team leader has tasked you with troubleshooting and resolving the issue.

Your task is to:

- Examine each of these issues and propose potential solutions.
- Break into smaller groups and discuss strategies on what solutions could be rendered for each of the identified issues.
- Develop an action plan for implementation.

A) Examine each of these issues and propose potential solutions.

i) High server loads during peak hours, slow response times, and intermittent downtimes.

- Use cloud services that can automatically add more power during busy times.
For example: Azure App Service (It offers automatic scaling based on demand).

ii) Slow data retrieval and increased server load due to unoptimized database queries.

- Add indexes to speed up search in the database.
- Show data in pages instead of loading everything at once.
- Move old data to another place, so the main database is faster.

iii) Rendering problems and decreased performance on certain browsers.

- Test the website on different browsers.

- Use standard code that works everywhere.
- Add backup options for features that don't work in all browsers.

B) Break into smaller groups and discuss strategies on what solutions could be rendered for each of the identified issues.

Group 1: Focus on server performance and load balancing.

- Research cloud solutions like Azure and caching strategies.

Group 2: Dive into database optimization.

- Analyze current queries, suggest improvements, and test performance.

Group 3: Handle frontend browser compatibility.

- Test different browsers, fix CSS, JS issues, and ensure responsiveness.
- My suggestion is a bootstrap.

C) Develop an action plan for implementation.

- Set up load balancer.
- Optimize database queries and add indexes.
- Conduct cross browser testing.